
Apex Trust Bank Plc

Fraud & Risk Intelligence Analysis

Project Documentation

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Role:

Data Analyst

Tools Used:

- Microsoft Excel
- MySQL
- Tableau

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1. Executive Summary

Financial fraud remains one of the biggest challenges facing banks today. Every day, thousands of transactions are processed across different channels, making it important to detect unusual activities early and reduce financial losses.

This project focuses on analyzing transaction data from Apex Trust Bank Plc to understand customer transaction patterns, identify fraudulent activities, and provide useful insights that can support better decision-making.

The project began with cleaning the raw dataset using MySQL to ensure the data was accurate and ready for analysis. After cleaning the data, Tableau was used to build an interactive dashboard that highlights important fraud trends, transaction patterns, customer behavior, and risk indicators.

The final dashboard provides a clear view of the bank's transaction activities through key performance indicators (KPIs), charts, and filters. It also presents business insights and recommendations that can help improve fraud monitoring and operational efficiency.

This project helped strengthen my skills in SQL, data cleaning, data visualization, and business analysis while showing the importance of preparing quality data before creating dashboards.

2. Business Background

Banks process thousands of financial transactions every day through channels such as mobile banking, internet banking, ATMs, POS terminals, USSD, and physical branches. While these services make banking more convenient, they also create opportunities for fraudulent activities.

To reduce financial losses and improve customer trust, banks need to monitor transactions continuously and identify suspicious activities as early as possible.

Apex Trust Bank provided transaction data that could be analyzed to understand customer behavior, identify fraud patterns, and measure different risk indicators. Before the analysis could begin, the data needed to be cleaned and validated to ensure reliable results.

The purpose of this project was not only to build a dashboard but also to transform raw transaction data into meaningful information that supports better business decisions.

3. Project Objectives

The main objectives of this project were to:

- Clean and prepare the banking transaction data using SQL.
- Identify fraudulent transactions within the dataset.
- Calculate important fraud and transaction KPIs.

- Analyze fraud trends over time.
 - Identify high-risk transaction channels.
 - Identify states with higher fraud activities.
 - Understand which transaction types are most affected by fraud.
 - Build an interactive Tableau dashboard for easy analysis.
 - Provide recommendations that can help reduce fraud risk.
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4. Business Questions

The project was designed to answer the following business questions:

- How many transactions were processed?
 - What is the total value of all transactions?
 - How many fraudulent transactions were recorded?
 - What is the fraud rate?
 - What is the average risk score?
 - Which transaction channels recorded the highest fraud cases?
 - Which transaction types are most associated with fraud?
 - Which states experienced the highest fraud activities?
 - How did fraud change over time?
 - What business actions can help reduce fraud risk?
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5. Dataset Overview

The dataset contains banking transaction records from Apex Trust Bank Plc covering customer information, account details, transaction activities, fraud indicators, and risk scores.

The dataset was used to analyze transaction behaviour and identify fraud patterns across different banking channels.

Dataset Highlights

- Approximately **2,000 transaction records**

- Data covers the period from **January 2024 to July 2026**
- Customer demographic information
- Transaction details
- Account information
- Banking channels
- Fraud indicators
- Risk scores
- Transaction status
- Customer location

Important Fields Used

Some of the important fields included:

- Transaction ID
- Customer ID
- Customer Name
- Customer Age
- Customer Gender
- Customer State
- Branch
- Account Type
- Transaction Date
- Transaction Type
- Transaction Channel
- Amount
- Transaction Status
- Fraud Flag
- Fraud Type

- Risk Score

These fields formed the foundation for the SQL data cleaning process and the Tableau dashboard.

Part 2: Data Cleaning and Dashboard Development

SQL Data Cleaning Process

Database Creation

The first step in this project was creating a new database in MySQL Workbench to store the banking transaction dataset. A dedicated database was created to keep the project organized and to ensure that all cleaning and analysis were performed in one environment.

After creating the database, the transaction table was imported successfully and checked to confirm that all records and columns had been loaded correctly.

Importing the Dataset

The dataset was imported from Excel into MySQL Workbench using the Table Data Import Wizard. After the import was completed, the number of rows and columns was verified to ensure that the data had been imported correctly.

Initial queries were also executed to understand the structure of the dataset and identify the available fields for analysis.

Data Exploration

Before making any changes, the dataset was explored to understand its contents.

Several SQL queries were used to:

- View the first few rows of the dataset.
- Identify all available columns.
- Check unique values within important fields.
- Understand transaction types, payment channels, customer locations, fraud categories, and transaction status.

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This exploration helped identify areas that required cleaning before the data could be used for visualization.

```
140     SET customer_age = NULL
141     WHERE age_flag = 'invalid_age';
142
143 •   SELECT MIN(customer_age), MAX(customer_age)
144     FROM apex_bank
145     WHERE customer_age IS NOT NULL;
146
147 •   SELECT COUNT(*)
148     FROM apex_bank
149     WHERE customer_gender IS NULL;
150
151 •   SELECT DISTINCT customer_gender
152     FROM apex_bank
153     WHERE customer_gender != TRIM(customer_gender);
154
155 •   SELECT DISTINCT customer_gender, COUNT(*)
```

SQL Data Exploration and Cleaning in MySQL Workbench

Duplicate Checks

One of the first cleaning tasks was checking for duplicate records.

Duplicate Transaction IDs were identified using SQL queries. After reviewing the results carefully, it was discovered that the records were not true duplicates because other transaction details differed.

As a result, no valid transaction records were removed, preventing unnecessary data loss.

Missing Values

The dataset contained missing values in several columns.

SQL queries were used to identify blank and NULL values across the dataset. Missing values were found in fields such as Merchant Category, Channel, Currency and some customer information.

Instead of deleting records, missing values were reviewed individually. Where appropriate, values such as "Unknown" or "Not Applicable" were used to maintain consistency while preserving the original data.

Standardizing Names

Some text fields contained inconsistent formatting.

Cleaning activities included:

- Removing unnecessary spaces from customer names.
- Standardizing customer state names.
- Making transaction categories consistent.
- Ensuring account types and currencies followed one naming convention.

This improved the accuracy of grouping and filtering during dashboard creation.

Date Formatting

The dataset contained date values that required standardization.

New cleaned date columns were created using SQL to ensure that transaction dates and account opening dates followed a consistent DATE format.

Using properly formatted dates made it easier to build trend analyses and monthly visualizations in Tableau.

Data Validation

After completing the cleaning process, validation checks were carried out to ensure data quality.

The following checks were performed:

- Verified record counts before and after cleaning.
- Confirmed that important fields contained valid values.
- Reviewed transaction status categories.
- Checked fraud indicators.

- Validated numeric fields such as transaction amount and account balance.

These checks ensured that the cleaned dataset was reliable and ready for visualization.

Challenges Encountered

During the cleaning process, several challenges were encountered.

Inconsistent Data

Some columns contained different spellings and formats for the same values, making analysis difficult.

Solution

The values were standardized using SQL functions to ensure consistency throughout the dataset.

Missing Information

Some records contained missing values in important fields.

Solution

The missing values were investigated individually. Where appropriate, they were labelled as "Unknown" or "Not Applicable" instead of deleting the records.

Duplicate Records

Some transaction IDs initially appeared to be duplicates.

Solution

After careful investigation, it was confirmed that these records represented different transactions rather than true duplicates, so they were retained.

Date Formatting Issues

Date fields were stored in inconsistent formats.

Solution

New cleaned date columns were created and converted into the correct SQL DATE format for easier reporting.

Lessons Learned During Data Cleaning

This project reinforced the importance of cleaning data before performing any analysis.

Some key lessons learned include:

- Never assume duplicate records should be deleted without investigation.
 - Always validate data after every cleaning step.
 - Standardized data improves reporting accuracy.
 - Data quality has a direct impact on dashboard performance and business decisions.
 - Careful data exploration helps uncover hidden issues early in the analysis process.
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Preparing the Cleaned Dataset for Tableau

Once the dataset had been cleaned and validated, it was exported from MySQL and saved as a clean Excel file for visualization in Tableau.

Before importing the file into Tableau, a final review confirmed that:

- All required columns were available.
 - Date fields were correctly formatted.
 - Numeric fields were suitable for calculations.
 - Text fields were standardized.
 - The dataset was ready for dashboard development.
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Dashboard Development Process

Why I Chose Tableau

Tableau was selected because it provides powerful visualization capabilities and makes it easy to create interactive dashboards.

It allows users to explore business data through filters, drill-down analysis, maps, trend charts, and KPI cards. These features make Tableau an excellent tool for fraud monitoring and transaction analysis.

Dashboard Planning

Before building the dashboard, the business questions were reviewed to determine which metrics and visualizations would provide the most value.

The dashboard was designed to answer questions such as:

- How many transactions occurred?
 - How much money was processed?
 - Which channels recorded the most fraud?
 - Which fraud types were most common?
 - Which states experienced higher fraud activity?
 - How has fraud changed over time?
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KPI Selection

The dashboard includes key performance indicators that provide an overview of transaction performance and fraud risk.

The selected KPIs include:

- Total Transactions
- Total Transaction Value
- Total Fraudulent Transactions
- Fraud Rate
- Average Risk Score
- Failed and Reversed Transactions

These KPIs provide decision-makers with a quick summary of business performance before exploring detailed charts.

Chart Selection

Different chart types were selected based on the type of information being presented.

The dashboard includes:

- Line Chart for fraud trends over time.
- Bar Chart for fraud by fraud type.
- Donut Charts for transaction types and payment channels.
- Map showing fraud distribution across customer states.

Each visualization was selected because it presents the data in a clear and meaningful way.

Filters

Interactive filters were added to improve user experience.

Users can filter the dashboard by:

- Transaction Status
- Channel
- Transaction Type
- Customer State
- Date

These filters allow users to focus on specific areas of interest without changing the underlying dashboard.

Layout and Design Decisions

A clean, modern dashboard layout was chosen to improve readability and make key business metrics easy to identify. A light background was combined with navy blue and complementary accent colors to create a professional appearance while maintaining strong visual contrast. KPI cards were placed at the top for quick access to key metrics, charts were arranged logically to guide users through the analysis, and interactive filters were positioned for easy navigation.

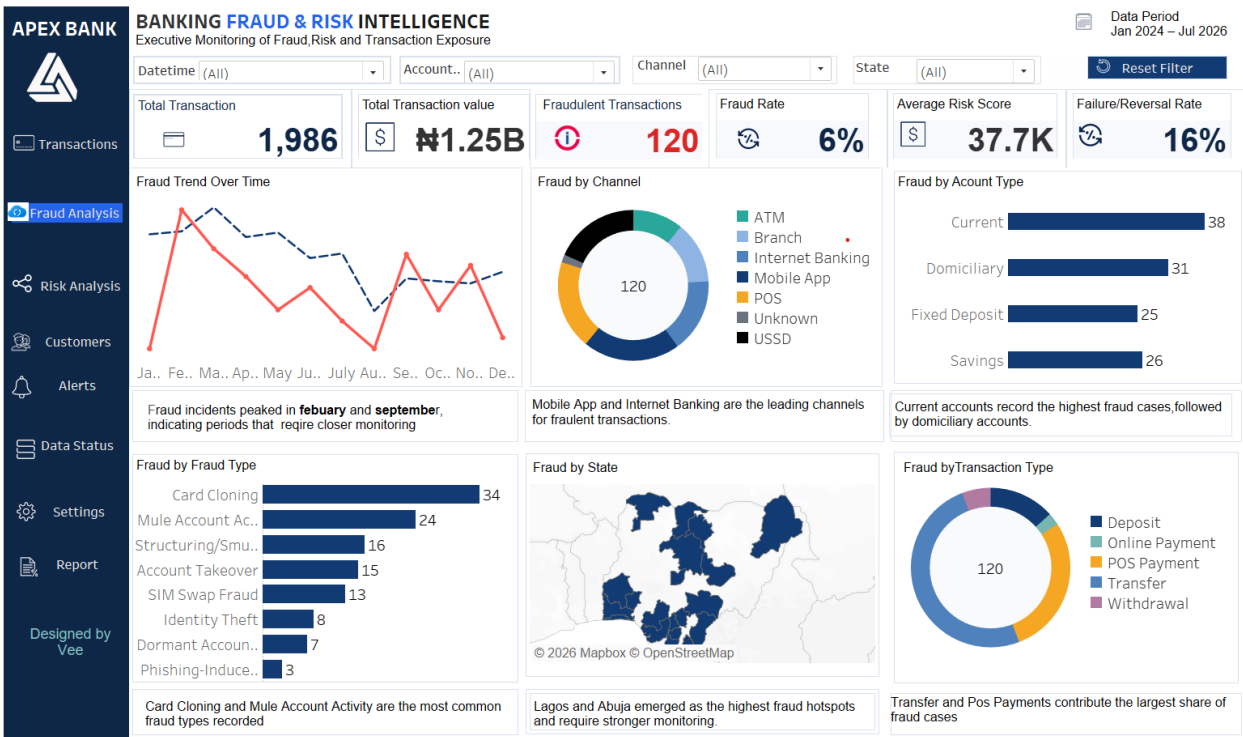
Consistent spacing, alignment, and color usage helped create a dashboard that is both visually appealing and user-friendly.

Dashboard Analysis, Business Insights & Conclusion

Dashboard Overview

The Tableau dashboard was designed to provide a clear overview of transaction activities and fraud risks within Apex Trust Bank. The dashboard allows decision-makers to quickly monitor fraud patterns, transaction performance, customer behavior, and operational risks through interactive visualizations and key performance indicators (KPIs).

By combining summary metrics with visual charts, the dashboard helps management identify unusual trends, evaluate transaction quality, and make informed decisions to improve fraud detection and customer security.



Final Tableau Fraud & Risk Intelligence Dashboard

Key Performance Indicators (KPIs)

1. Total Transactions

This KPI shows the total number of transactions processed during the analysis period.

Business Value

- Measures overall banking activity.
 - Helps management understand transaction volume.
 - Serves as the baseline for comparing fraud and failure rates.
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2. Total Transaction Value

This KPI displays the total monetary value of all transactions processed.

Business Value

- Shows the overall amount moving through the banking system.
 - Helps evaluate customer activity.
 - Indicates the financial exposure of the bank.
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3. Total Fraudulent Transactions

This KPI represents the number of transactions flagged as fraudulent.

Business Value

- Measures the extent of fraud within the system.
 - Helps management monitor fraud levels.
 - Supports fraud prevention strategies.
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4. Fraud Rate (%)

This KPI calculates the percentage of transactions identified as fraudulent.

Business Value

- Shows how common fraud is compared to all transactions.

- Makes it easier to compare fraud performance across different periods.
 - Acts as an important fraud monitoring metric.
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5. Average Risk Score

This KPI displays the average fraud risk score assigned to all transactions.

Business Value

- Measures the overall fraud risk within the transaction dataset.
 - Helps identify whether customer activities are becoming riskier.
 - Supports proactive monitoring before fraud occurs.
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6. Transaction Failure & Reversal Rate

This KPI combines failed and reversed transactions.

Business Value

- Measures operational efficiency.
 - Identifies possible system issues.
 - Highlights customer experience problems that may require investigation.
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Dashboard Charts

Fraud by Fraud Type

This chart compares different categories of fraud detected in the bank.

Insight

Some fraud categories occur much more frequently than others, allowing the bank to identify the most common fraud methods being used by criminals.

Fraud by Transaction Type

This chart shows which transaction types experience the highest number of fraudulent activities.

Insight

Certain transaction types carry higher fraud risk than others. This information allows management to strengthen monitoring on those specific transaction types.

Fraud Trend Over Time

This line chart tracks fraudulent transactions across the reporting period.

Insight

The trend helps identify periods where fraud activity increased or decreased. Any sudden spike may indicate coordinated fraud attacks or changes in customer behavior.

Fraud by Channel

This visualization compares fraud across banking channels such as Mobile Banking, ATM, POS, USSD, Internet Banking, and Branch transactions.

Insight

Some channels experience higher fraud occurrences than others, helping management prioritize security improvements where they are needed most.

Fraud Distribution by State

The map visualization highlights where fraudulent transactions occur across different states.

Insight

Certain states report higher fraud activity than others, allowing the bank to allocate investigation resources more effectively and strengthen regional monitoring.

Key Business Insights

After analyzing the dataset, several important findings were observed:

- Fraud is concentrated within specific transaction types rather than being evenly distributed.

- Certain banking channels experience significantly higher fraud rates, suggesting stronger security controls are needed.
 - Fraud activity fluctuates over time, indicating the importance of continuous monitoring rather than periodic reviews.
 - Geographic analysis shows that fraud is more concentrated in some states, enabling targeted risk management.
 - Failed and reversed transactions represent operational issues that may affect customer satisfaction if left unresolved.
 - Monitoring average risk scores alongside fraud counts provides an early warning system for emerging threats.
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Recommendations

Based on the findings from this analysis, the following recommendations are proposed:

1. Increase fraud monitoring on high-risk transaction types and banking channels.
 2. Deploy automated alerts for transactions with unusually high risk scores.
 3. Conduct periodic reviews of states with consistently high fraud activity.
 4. Improve transaction validation processes to reduce failed and reversed transactions.
 5. Continue monitoring fraud trends through an interactive dashboard to support timely decision-making.
 6. Update fraud detection rules regularly to adapt to changing fraud patterns.
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Project Reflection

This project provided valuable experience in transforming raw banking transaction data into meaningful business insights. The process involved cleaning inconsistent data using SQL, preparing the dataset for analysis, and designing an interactive Tableau dashboard that communicates fraud risks clearly.

Throughout the project, I strengthened my skills in SQL data cleaning, data validation, Tableau dashboard development, KPI design, data visualization, and business storytelling. Most importantly, I learned that effective analytics is not just about creating attractive dashboards but about helping decision-makers understand problems and take informed actions.

Conclusion

The Apex Trust Bank Fraud & Risk Analysis Dashboard demonstrates how data analytics can support fraud detection and operational decision-making in the banking sector. By combining SQL for data preparation with Tableau for visualization, the project delivers a clear and interactive view of transaction performance, fraud patterns, operational risks, and geographic trends.

The dashboard enables stakeholders to monitor key metrics, identify areas of concern, and make data-driven decisions that improve fraud prevention, operational efficiency, and customer confidence. This project showcases the practical application of business intelligence tools to solve real-world financial challenges and reflects a complete analytics workflow—from raw data to actionable insights.